LECTURE 1

Topics

Capital preservation, supply of financial assets, investment process and financial markets

- The income that a person earns changes over time
  - Predictable: wage increases, family inheritance, etc.
  - Unpredictable: layoffs, new opportunities, winning the lottery, etc.
- The income earned over a period of time is typically used to finance consumption needs over that period of time
- Consumption needs can change over time (e.g., education)
- What if the income earned over a certain period is not enough to finance our consumption needs for that period?
  1. Reduce your consumption needs
  2. Borrow against future income
  3. Save/invest when income is high for when income is low

- Reducing consumption needs can be very costly, e.g lowering social status
- Borrowing against future income may require collateral
- Saving/investing when possible seems most natural
- How exactly do we save/invest our excess cash?
  - Hold on to it, e.g., checking account with no interest/costs
  - Acquire an asset that generates cash flows in the future
- If all acquisitions carry risk and/or costs, then some consumers may prefer to hold on to the cash
- However, if there is a risk-free and zero-cost term deposit that pays \$1 + interest for each \$1 deposited, then some consumers will prefer acquiring this asset to holding on to the cash

- A consumer that has some extra cash today could hold on to its cash and have \$1 tomorrow for each \$1 today
- Alternatively, it could acquire a risk-free zero-cost term deposit and have \$1 tomorrow + \$ interest tomorrow for each \$1 today
- The second alternative gives the consumer more purchasing power tomorrow
- Thus, all consumers prefer the second alternative
- Effectively, \$1 today is worth \$1 tomorrow + \$ interest tomorrow
- This is known as the time value of money (TVM)

TVM

\$1 today = \$1 tomorrow + \$ interest tomorrow > \$1 tomorrow,
or, in short:
\$1 today > \$1 tomorrow

- In the US, banks offer time deposits that are federally insured (up to $250,000) and thus, risk-free
- Many of these insured time deposits have virtually zero costs
- Thus, in the US, risk-free zero-cost time deposits exist

- Of course, a US consumer that wants to save its extra cash is not constraint to risk-free zero-cost time deposits only
- If this consumer is willing to take on some risk, it could invest in assets that could generate higher, albeit riskier, payoffs than the risk-free zero-cost time deposit
- These risky assets can be of two types:
  - Real assets, such as a producing farm, a gold mine, the assets of a firm producing goods or services, etc.
  - Financial assets, such as the equity or the debt issued by a firm, the debt issued by a country, mutual funds, etc.

- Thus, a consumer who wants to save/invest can either:
  - Save, by acquiring the risk-free zero-cost term deposit, or
  - Invest, by acquiring one or more real and financial assets
- At this point the following two questions become relevant:
  - How much should the consumer save vs how much it should invest?
  - And if it chooses to invest, which assets should it choose and how much should it acquire of each asset?
- These two questions are collectively known as the consumer's optimal portfolio selection or optimal asset allocation

Optimal asset allocation

A consumer's optimal asset allocation is a set of decisions regarding how much to save and invest and in what assets

- To address the consumer's optimal asset allocation problem we will need some inputs:
    1. Consumer's objective ("optimal" means that this objective is achieved)
    2. The interest rate associated with the risk-free zero-cost savings opportunity
    3. The investment opportunity set of available real and financial assets
    4. The set of possible future cash flows generated by each asset and their corresponding likelihoods (also known as the "risk profile" of an asset)

- In real life, a consumer's objective can be quite specific: a downpayment to buy a house, kids' college expenses, etc.
- Many of these real-life objectives are in fact identifying predictable future consumption needs:
  - More housing services (bigger house to accommodate a growing family)
  - More education services (your kids may want to go to college), etc.
- From an economics perspective, all these seemingly different real-life objectives are part of one overarching objective:

Consumer's objective

Maximize the life-time happiness that the consumer derives from fulfilling current and future consumption needs

- In economics we have specific ways to quantify the "happiness" associated with fulfilling a certain consumption need

- The investment opportunity set consists of all available real and financial assets
- Some real assets are readily available and can be acquired:
  - a real estate property (e.g., developed or undeveloped land)
  - a copper mine
  - the assets of a firm
- Other real assets are not readily available and may not be obvious to acquire:
  - Exploiting a recent innovation to create a new market
- Financial assets are ultimately claims on a firm's real assets
- Financial assets associated with a public firm (i.e., listed on an exchange) are usually well publicized and easy to acquire
- Financial assets associated with private firms are usually not publicized and may be more difficult to acquire

- A consumer's decision to invest in an asset will depend on the risk profile of the asset, which consists of:
  - The possible future cash flows generated by the asset
  - The likelihood (or probability) associated with each possible cash flow
- To put together the risk profile of an asset we need information
- We generally rely on historical data on the asset or on similar assets to tease out the necessary information
- An important aspect of this process is the use of traditional and modern statistical methods
- Generally, we have far more historical data for financial assets than for real assets
- Much of our focus on the optimal asset allocation problem will be on financial assets with available historical data

- A firm is a portfolio of real assets (also known as "projects")
- For example, Ford is a portfolio of projects that are each associated with the production and sale of a Ford vehicle model
- Ford projects require large capital investments (i.e., they are "capital intensive"), which have to be financed
- To finance its projects, Ford sells ("issues") standardized securities that are ultimately claims on Ford's projects
    - A buyer of a Ford security is entitled to receive a portion of all future cash flows generated by Ford's projects
    - In exchange, the buyer must pay for the security upfront
- Ford uses these upfront payments to finance its projects
- The cash flows generated by Ford's projects are then used to repay investors according to their respective claims

- What type of securities can a firm issue?
- Debt securities:
  1. Corporate bonds are issued by public firms and promise fixed periodic payments ("coupons") until expiration
  2. Corporate bonds issued by certain federal agencies (e.g., Fannie Mae) are known as agency bonds
  3. Mortgage-backed securities (MBS) and asset-backed securities (ABS) are corporate bonds backed by pools of mortgage loans and other loans, respectively
  4. Eurobonds are corporate bonds issued by domestic firms with coupon payments denominated in a foreign currency
  5. Yankee bonds are dollar-denominated corporate bonds issued by foreign firms in the US
  6. Commercial paper is a type of corporate bond with very short maturity issued typically by finance firms (e.g., banks)

- In the event that the firm defaults on its debt obligations, debt holders have a claim on the firm's assets
- Certain such claims have to be fulfilled before others (we say that the former claims have "priority" over the others)
- Senior bonds have priority over junior bonds in the event of issuer default
- We sometimes say that the claim of a junior bond is subordinated to that of a senior bond
- Corporate bonds are generally a form of non-collateralized (or "unsecured") debt
- Certain corporate bonds can be collateralized by a portfolio of assets ("collateral"), such that if the issuer defaults, the bondholders get the collateral

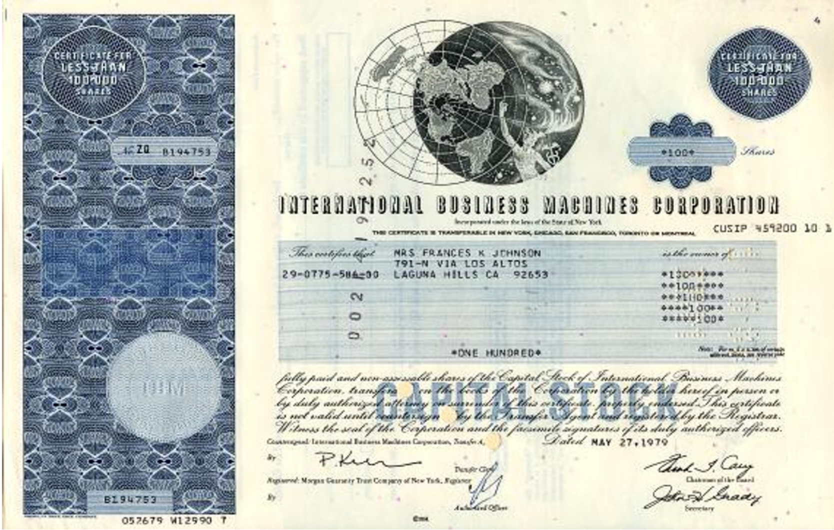
- Equity securities:
  1. Common stock of a public US firm entitles the buyer ("shareholder") to the residual claim on the firm's futures cash flows after payments on other securities have been serviced
  2. American Depository Receipts ("ADR") are certificates that trade on the US stock markets and that represent ownership in the common stock of a foreign public firm

  - Shareholders may receive dividends from the firm
  - Common stock is the only security type that comes with voting rights, effectively making shareholders the owners of the firm
  - An ADR certificate may not necessarily correspond to exactly one foreign share of the foreign firm

- Hybrid securities:
  1. Preferred stock that promise a fixed dividend payment as long as the firm distributes dividends
  2. Convertible bond is similar to a corporate bond but with the option to convert a bond for a predetermined number of common stock shares
  3. Convertible preferred stock is similar to preferred stock but with the option to convert a share of preferred stock for a predetermined number of common stock shares

  - Preferred stock holders receives dividend payments before common stock holders
  - If a firm suspends its dividend distribution, neither preferred stock nor common stock holders receive dividends
  - In the event that the firm defaults on its debt, the claims of the preferred stock holders have priority over the claims of the common stock holders

- Certificate of common stock ownership in IBM



- Certificate of corporate bond ownership in IBM

- Certificate of preferred stock ownership in Fisher Body Ohio Co.

- Most countries, states, counties and municipalities finance their budget needs using debt securities

    1. US Treasury Bills, Notes and Bonds are, respectively, short-term (up to 1 year), medium-term (up to 10 years), and long-term bonds (up to 30 years) issued by the US Government (US Treasury)
    2. Municipal bonds ("Munis") are bonds issued by a US local government such as state, county, or municipalities
        - General obligation munis are used to finance the budget needs of a local government
        - Revenue munis are used to finance a specific project and its coupon payments are made from the project's revenues
    3. Sovereign bonds are bonds issued by a foreign country, and its coupon payments could be denominated in the currency of that country

- Finally, consumers could also invest in complex financial instruments that make payments based on the cash flows of one of the standardized financial assets that we discussed above
- These complex financial assets are called derivatives
- Unlike any of the standardized financial assets discussed above, derivatives are not supplied by firms or governments
- A derivative is a contract between two consumers, who agree to make payments to each other based on the payoffs of a standardized financial assets (e.g., the stock of a firm)
- Given that any two consumers can enter a derivative contract, the size of the derivatives market (all derivatives contracts that are still outstanding) is far larger than the size of the market for any of the standardized financial assets

- Example of derivatives include:
  1. Stock options that give the buyer the right to buy or sell the stock of a firm at a predetermined price at some future time
  2. Commodity (e.g., gold) and currency (e.g., Euros) futures that allow the buyer to purchase a certain amount of commodity or foreign currency, respectively, at a predetermined price at a future date
  3. Interest rate swaps that allow the buyer to exchange floating interest payments for fixed interest payments
  4. Credit default swaps that allow the buyer to make quarterly payments and in exchange receive compensation in the event that a firm or government defaults on its debt
- Derivatives are customized to fit the needs of their buyers, and there's usually a large variety of derivative types

- Finance firms such as banks can borrow or lend over very short periods of time (e.g., overnight) using repurchase agreements
- In a repurchase agreement ("repo") a finance firm "sells" a portfolio of assets to another firm with an agreement to repurchase them back at a slightly higher price after 24 hours
    1. Term repurchase agreements specify that the portfolio of assets have to be repurchased after 24 hours
    2. Open repurchase agreements do not have a set repurchase date, and effectively roll over for the next 24 hours until one of the parties calls for termination (and repurchase)
- The portfolio of assets in a repo consists of high quality bonds
- The US government also makes use of repos when they buy/sell US Treasury bonds to change the target federal fund rate

- As noted above, there are quite a number of financial assets
- Consumers will eventually have to decide how much to invest in each of these assets
- However, some of these financial assets share similarities and can be thought of collectively as being part of the same asset class
- Classifying financial assets into asset classes simplifies the economics of consumer's investment decision

- We identify the following asset classes:
  1. Money-like instruments, such as US Treasury Bills, Commercial paper, Repurchase agreements
  2. US Treasury Notes and Bonds
  3. Domestic corporate bonds
  4. Municipal bonds
  5. Real estate (e.g., agency bonds, home ownership)
  6. Sovereign bonds issued by foreign governments
  7. Foreign corporate bonds denominated in US dollars or foreign currency
  8. Domestic equity
  9. ADR certificates and international equity
  10. Commodity and currency (e.g., futures)
  11. Other derivatives (e.g., stock options, and swaps)

- Aggregate US consumer's holdings of various asset classes:

Assets	$ Billion	% Total	Liabilities and Net Worth	$ Billion	% Total
Real assets			**Liabilities**		
Real estate	$ 37,558	24.4%	Mortgages	$ 11,331	7.4%
Consumer durables	6362	4.1%	Consumer credit	4,163	2.7%
Other	679	0.4%	Bank and other loans	1,125	0.7%
Total real assets	$ 44,599	28.9%	Other	624	0.4%
			Total liabilities	$ 17,244	11.2%
Financial assets					
Deposits and money market shares	$ 17,374	11.3%			
Life insurance reserves	1,854	1.2%			
Pension reserves	29,876	19.4%			
Corporate equity	28,285	18.3%			
Equity in noncorp. business	13,114	8.5%			
Mutual fund shares	11,661	7.6%			
Debt securities	5,772	3.7%			
Other	1,626	1.1%			
Total financial assets	$109,562	71.1%	*Net worth*	136,917	88.8%
TOTAL	$154,161	100.0%		$154,161	100.0%

Table 1.1

Balance sheet of U.S. households

Note: Column sums may differ from total because of rounding error.

Source: *Flow of Funds Accounts of the United States*, Board of Governors of the Federal Reserve System, June 2021.

- Consumers' contribution to US aggregate assets

Table 1.2

Domestic net worth

Assets	$ Billion
Commercial real estate	$20,842.9
Residential real estate	45,816.3
Equipment & intellectual property	10,316.2
Inventories	2,944.5
Consumer durables	6,361.9
TOTAL	$86,282

Note: Column sums may differ from total because of rounding error.
Source: *Flow of Funds Accounts of the United States*, Board of Governors of the Federal Reserve System, June 2021.

- Consumer can choose to invest their capital themselves (self-directed investment) or delegate the investment decision to an agent (financial broker or advisor)
- There are generally two approaches to investing
- Top-down approach:
  - First, determine how much capital is to be allocated to each asset classes
  - Then, determine which particular securities to be held in each asset class
- Bottom-up approach
  - Investment based on attractively priced securities without as much concern for asset allocation

- Most of the financial assets introduced above trade on specialized markets
- Financial assets that are short term, low risk, and highly-liquid trade on the so-called Money Market
- Examples of such assets include:
  - US Treasury Bills
  - Commercial paper
  - Repos
- The interest rate benchmarks on the Money Market are usually used to "index" variable interest rate products, such as adjustable rate mortgage loans (or ARM)
- The main benchmark indexes relevant for the US market: LIBOR and SOFR

- London interbank offer rate (LIBOR) was the primary interest rate benchmark until 2021
  - Based on surveyed interbank lending interest rate quotes (not actual transactions)
  - Quite volatile and easy to manipulate by the banks providing the quotes (2012 scandal)
- Secured Overnight Financing Rate (SOFR)
  - New benchmark that replaced LIBOR in the US
  - Based on rates on overnight repurchase agreements collateralized by Treasury securities
- The difference between either of these benchmark indexes and the US Treasury Bills is viewed as a gauge of credit risk in the lending markets

- The TED Spread = Difference between LIBOR and US T-Bills

- The grey areas indicate financial crises

- Financial assets that are riskier and longer term than money market securities trade on Capital Markets
- Capital market are divided into 4 segments:
  - Equity markets (e.g., domestic equity securities and ADRs)
  - Bond markets (US Treasury Bonds and Notes, corporate bonds, agency bonds, munis)
  - Futures markets (e.g., commodities and currencies)
  - Options markets (e.g., stock options)

- In this class, whenever we refer to "market", we generally mean a benchmark index that tracks the US equity market
- The most popular indexes are:
  - Dow Jones Industrial Average (DJIA Index) tracks 30 large "blue-chip" US corporations
  - Standard and Poor's 500 (S&P 500 Index) tracks the largest 500 firms in the US
  - Russell 3000 Index tracks the largest 3000 stocks in the US (about 95% of the equity market)

- In the US the main equity markets are NYSE and NASDAQ
- New York Stock Exchange (NYSE)
  - Largest U.S. stock exchange, as measure by market value of listed stocks
  - Traditionally it used to be a broker/specialist system, where specialists made a market in a particular stock ("market makers") by trading with brokers
  - Investors need a broker to make a trade
  - Today this broker/specialist system is complemented by an electronic system that matches buy orders with sell orders automatically
  - NYSE is one of the few markets that can be fast and efficient at matching orders through its electronic trading system, yet still maintain the advantages of human interaction for complex trades

- National Association of Securities Dealers Automated Quotations (NASDAQ)
  - Consisted of a network of dealers that were primarily providing quotes, that were eventually connected via an electronic trading platform
  - The first electronic market in the world, where buy and sell orders from different dealers were matched automatically
  - Historically it tended to attract mainly technology stocks
  - Currently, the most active market in the US by volume

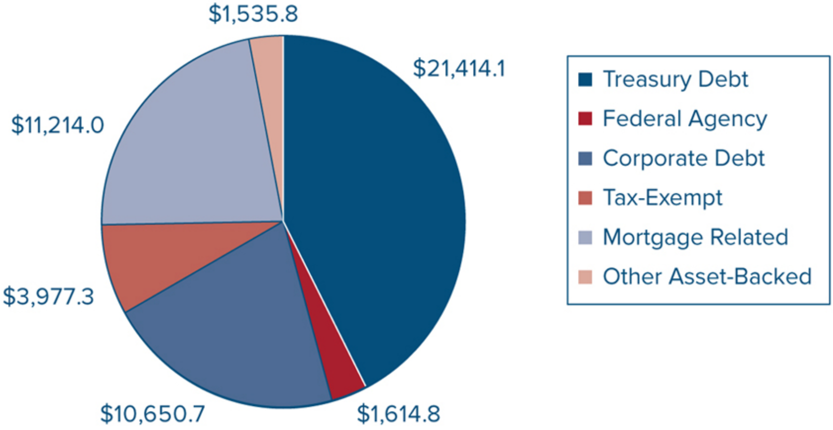
- In addition to NYSE and NASDAQ, stock trading in the US also takes place through alternative trading systems (ATS) such as ECNs
- Electronic Communication Networks (ECNs) are computer-operated trading networks that operate outside traditional stock exchanges ("off-the-floor trading")
- Their competitive advantage is speed of execution
- ECNs usually facilitate trades to members only ("closed book"), but there is a trend towards an "open book" model with no broker intermediaries
- Investors can access an ECN via a broker/dealer that is a member of the ECN or directly, if the ECN allows it
- ECNs allow members to trade outside traditional trading hours
- About 40% of stock trading in the US happens through ECNs

- Another important form of ATS is a Dark Pool
- A Dark Pool is a private exchange system that allow members to trade large blocks of stocks with minimal impact on the price of the stocks being traded
- Effectively the public is kept in the dark about a large block trade until the trade is realized
- Dark pools are available exclusively to institutional investors only
- Many of the largest banks have a dark pool platform for their institutional clients
- Hight-frequency trading relies on access to dark pools

- Electronic exchanges that provide for efficient stock trading have also emerged in other markets, such as futures and option markets
- The key feature that makes a financial asset amenable to trading on an electronic exchange is standardization
- Certain type of commodity and currency futures are sufficiently standardized that they can trade on exchanges
- Main futures exchanges in the US are the CME Group and ICE
- Main options exchange in the US is CBOE

- Where standardization of a financial asset is not possible, the only viable alternative for trading these financial assets is on over-the-counter (OTC) markets
- An OTC market is an informal networks of brokers and dealers where certain financial assets can be traded
- The bond market is an OTC market
- To buy a bond, an investor would have to contact a dealer who wouldthen locate the bond by calling other dealers
- Bonds are hard to standardize because a single company may have dozens of outstanding bond issues, differing by coupon, maturity, and seniority
- Relative to electronic exchanges, OTC markets are more illiquid (small trading volume in a specific asset) and have higher transaction costs

- The composition of the bond market



- Many futures, options and swaps cannot be standardized and therefore can only trade on OTC derivatives markets
- The main advantage of derivatives OTC markets over exchanges is the ability to produce bespoke derivative securities that fit perfectly the needs of an investor ("customization")
- Central to a well-function OTC market is the network of dealers that intermediate trade between investors
- These dealers are usually associated with large investment banks, and we will talk about them in Lecture 2